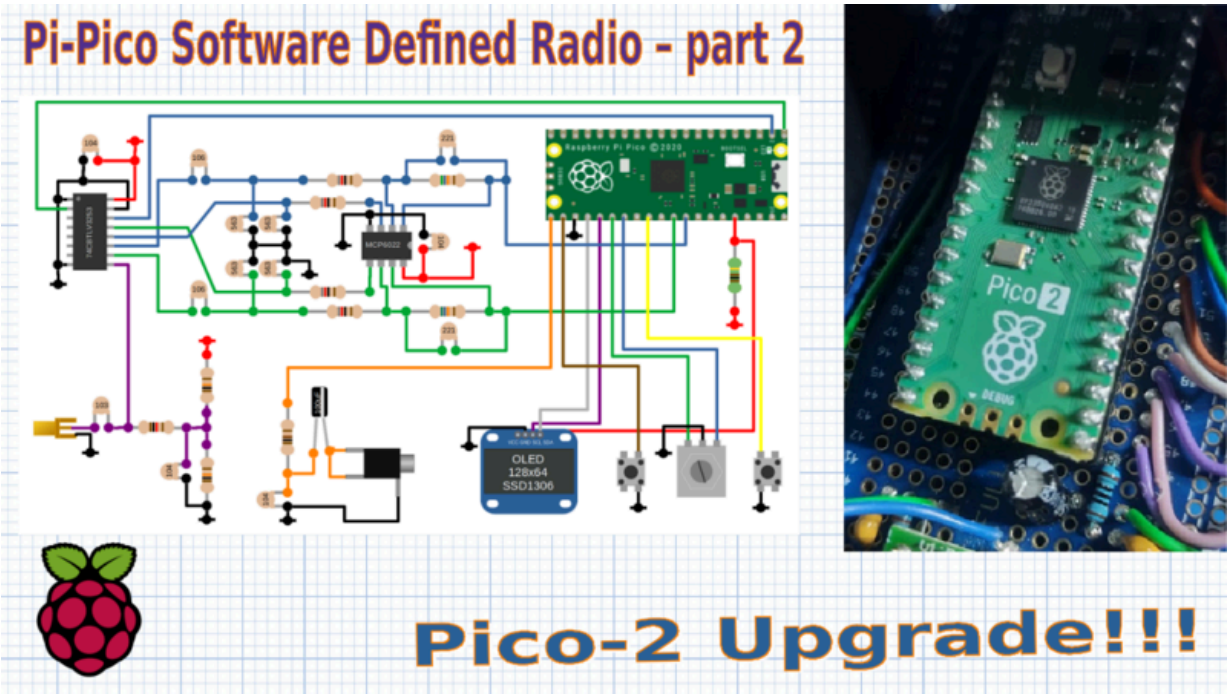
Breadboard Radio - Part 2



Exploring the Pi Pico 2

I recently received my Pi Pico 2 and was able to integrate it into the SDR with minimal code changes. Despite initial expectations of only marginal improvements, the performance boost was impressive.

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The key enhancements in the Pi Pico 2 are the Floating Point Unit (FPU) and the DSP co-processor. Although the SDR code is written in fixed-point arithmetic, the faster clock frequency of the Pico 2 provides a significant reduction in CPU usage—from around 80% on the original Pico to approximately 40% on the Pico 2.

The Pico 2 offers a choice of processors: the ARM Cortex M33 or the RISC-V Hazard 3. I tested both and found their performance similar in this fixed-point application. The success of the RISC-V processor suggests that we may see more of these processors in future projects.

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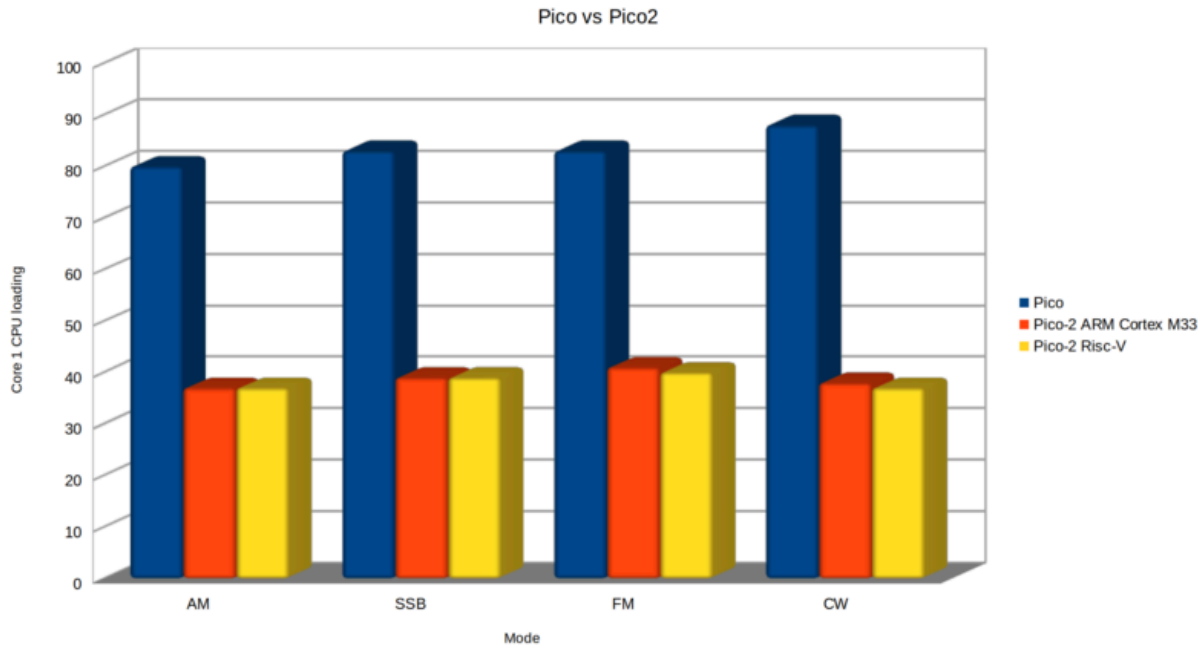
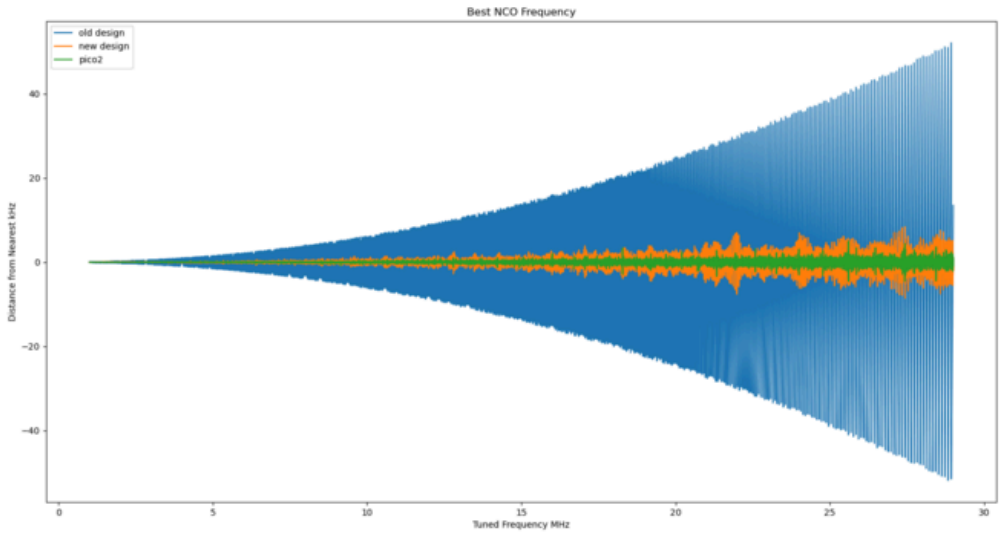


One curiosity was whether fixing the well-known ADC bug in the Pico would affect the SDR's performance. After testing the noise floor across various bands, I found no measurable or noticeable difference. Given that the SDR averages hundreds of samples, any occasional bad ADC readings likely get lost in the noise.



The increased clock frequency of the Pi Pico 2 also allows for finer resolution of the local oscillator, enhancing the SDR's performance, particularly on higher frequency bands. While the improvement is modest—just a couple of kHz—it contributes to a more predictable performance on the high bands.

Receiving Weather FAX

I received a comment asking if the SDR could be used to receive weather FAX. Although I hadn't tried this before, I gave it a go using a sound card and Fldigi on a PC. The setup worked without any issues, successfully downloading weather maps. This is a really interesting technical solution, it is quite impressive that a narrow band HF channel can be used to transmit FAX data, but it is also a testament to the skill of the people who can read and interpret these maps.

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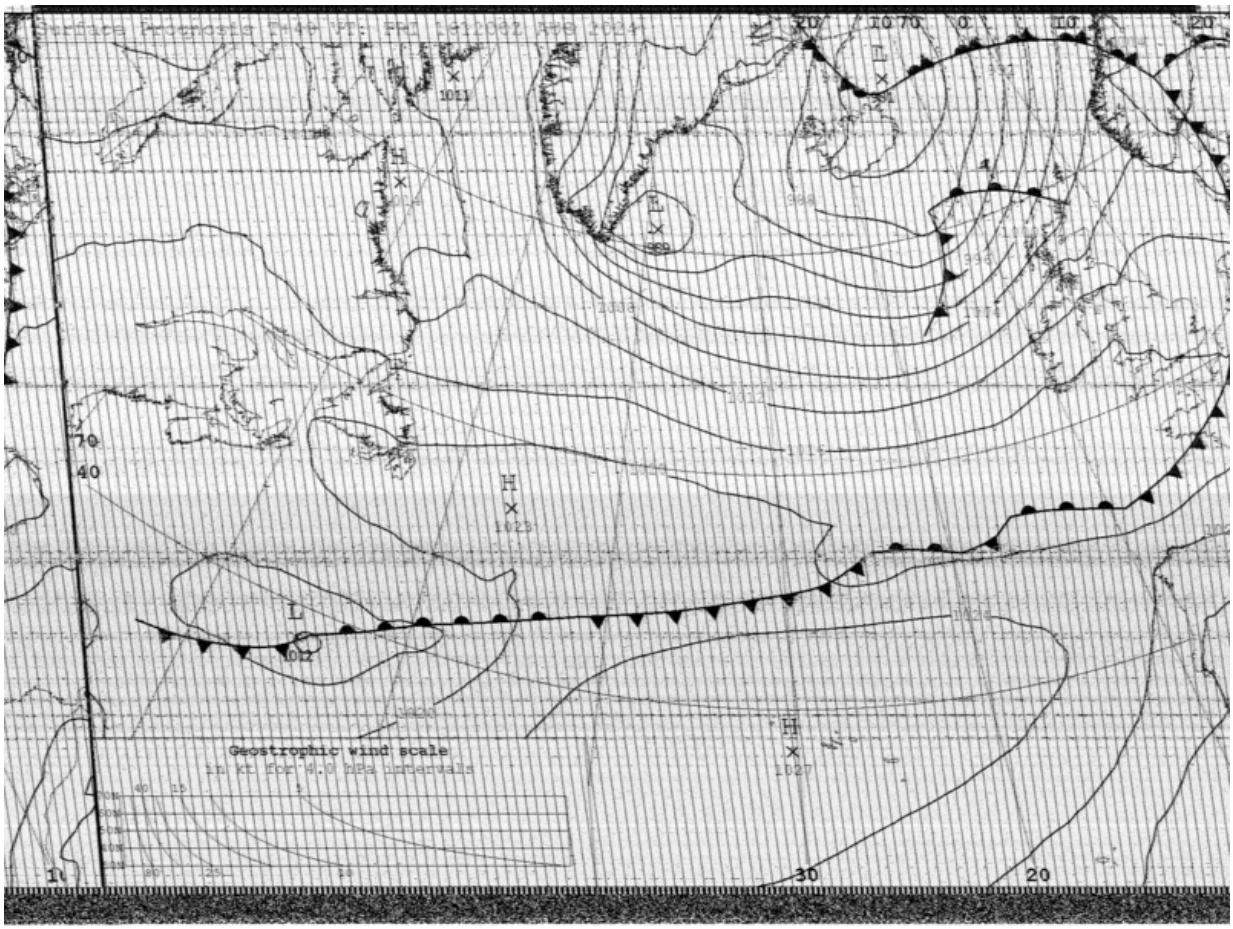

Operational Amplifier Alternatives

Component availability, particularly for op-amps, has been another concern. Fortunately, the op-amps used in this project are not particularly special, and suitable substitutes are widely available. Here are the key specifications to look for in an op-amp for this SDR.

Operating Voltage

The design uses the 3.3V output from the pico to drive the Tayloe detector. This is mainly so that the voltage stays constant regardless of the battery level. It also means that we don't have to worry about overdriving the ADC which only works at up to 3.3V. A starting point is to select a dual output with a minimum supply voltage of 3V or less.

Gain-Bandwidth Product:

A product of around 10 MHz is sufficient, given the reduced bandwidth of the detector (12 kHz) and a gain of 600.

Noise Performance:

Dan Tayloe's paper provides some guidance on the noise specification and presents a formula to calculate the Minimum Detectable Signal based on the op-amp performance.

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Sample Noise Calculations

Between the noise equation, and the signal equation, the input signal level necessary to produce an output signal equal to the output noise can be found:

0.9*S*4*G = G*Sqrt((4*0.9*0.45)**2 + 2*(Np)**2)*Sqrt(Bw)

Or

S = Sqrt((4*0.9*0.45)**2 + 2*(Np)**2)*Sqrt(Bw) / (0.9*4)

It is instructive to compare the system input noise to the signal level required to reach a 3 db S+N/N ratio. Such a table is shown below for a receiver with a 1000 Hz bandwidth.

System Noise @ Bw=1000Hz	Preamplifier Noise (nV/SqrtHz)	Signal level for 3 db S+N/N
14.2 nV / -143.8 dbm	9	113 nV / -125.8 dbm
14.2 nV / -143.8 dbm	4	55.3 nV / -132.0 dbm
14.2 nV / -143.8 dbm	2	34.8 nV / -136.0 dbm
14.2 nV / -143.8 dbm	1	27.3 nV / -138.1 dbm
14.2 nV / -143.8 dbm	0.8	17.3 nV / -142.1 dbm

At HF frequencies there is a great deal of noise on the bands, RECOMMENDATION ITU-R P.372-7 tells us the expected level of man-made noise at different frequencies.

5 Man-made noise

Median values of man-made noise power for a number of environments are shown in Fig. 10. The Figure also includes a curve for galactic noise (see § 6).

In all cases results are consistent with a linear variation of the median value, F_{am}, with frequency f of the form:

$$F_{am} = c - d \log f \qquad (11)$$

With f expressed in MHz, c and d take the values given in Table 1. Note that equation (11) is valid in the range 0.3 to 250 MHz for all the environmental categories except those of curves D and E as indicated on the Figure.

TABLE 1

Values of the constants *c* and *d*

Environmental category	*c*	*d*
Business (curve A)	76.8	27.7
Residential (curve B)	72.5	27.7
Rural (curve C)	67.2	27.7
Quiet rural (curve D)	53.6	28.6
Galactic noise (curve E)	52.0	23.0

The plot compares the expected receiver performance with the levels of band noise. We can see that an amplifier with a noise density of 9 nV/√Hz should offer comparable performance to a typical receiver, and would not limit the performance of the receiver under most circumstances.

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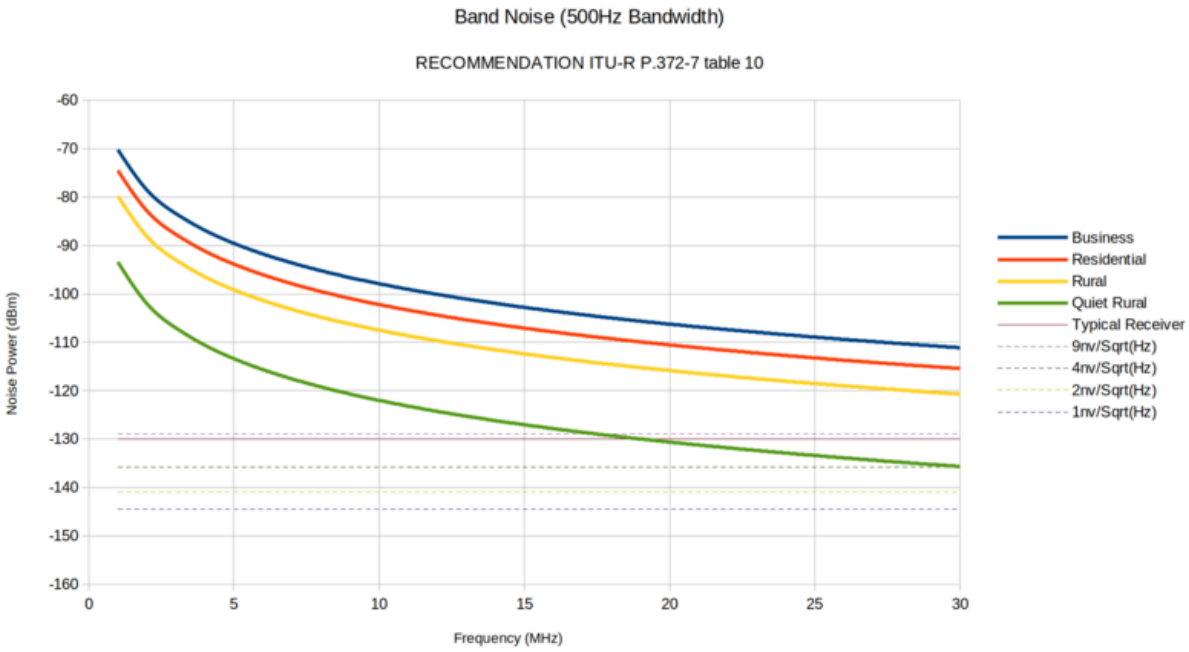

I selected a few of the cheapest operational amplifiers that meet this specification and tested them in the receiver.

	GBP (MHz)	Noise Vol age Density (nV/√Hz)	Cost (GBP)	DIP package
MCP6292	10	8.7	0.82	Yes
MCP6022	10	8.7	1.44	Yes
MCP662	60	6.8	1.25	No
OPA2607	50	3.8	1.26	No
OPA1662	22	3.3	1.44	No
LT6231	215	1.1	6.25	No
LTC6227	420	1	7.53	No

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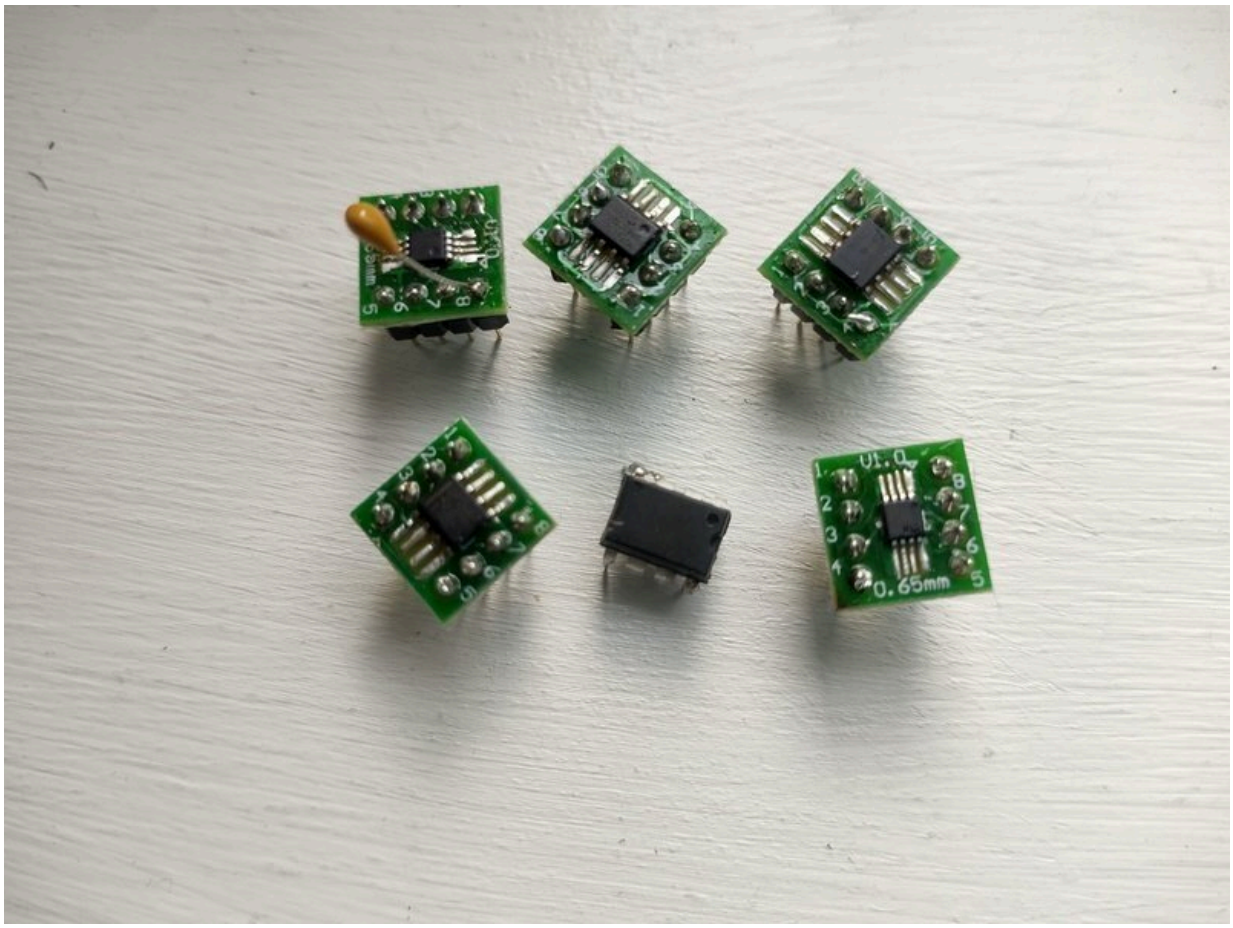

All of these devices were tested and worked without any issues. There was no noticeable difference in performance, so it isn't worth using expensive devices in this design. If you struggle to find the right op-amp, consider adapting the design to work with 5V devices, which broadens the range of available components.

Additional Improvements

I've also explored adding an external amplifier and speaker to the SDR. While low-cost PC speakers work well, a built-in speaker could be more convenient, especially for portable use. There are various low-power amplifier options available, allowing you to tailor the setup to your specific needs and budget.

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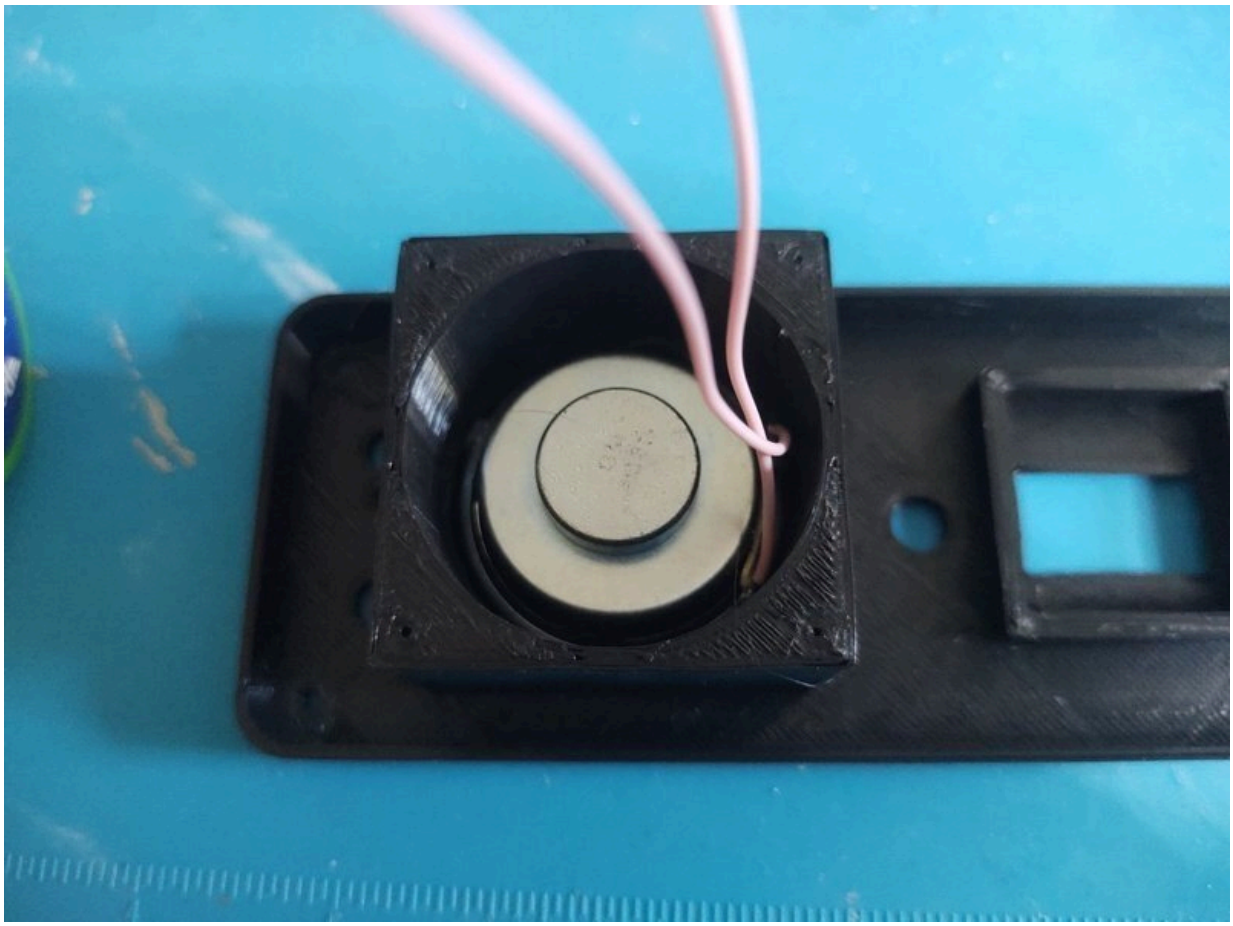

I have opted to use an inexpensive PAM8403 module. These can be obtained cheaply and can provide up to 3W output into a 4 ohm load. It uses a class-D design which allows efficiencies of up to 90% which makes it ideal for portable applications. As supplied, the module uses 10k ohm input resistors, this sets the gain much too high for this application, so I have replaced these with 100k ohm resistors to give a gain of around 2x.

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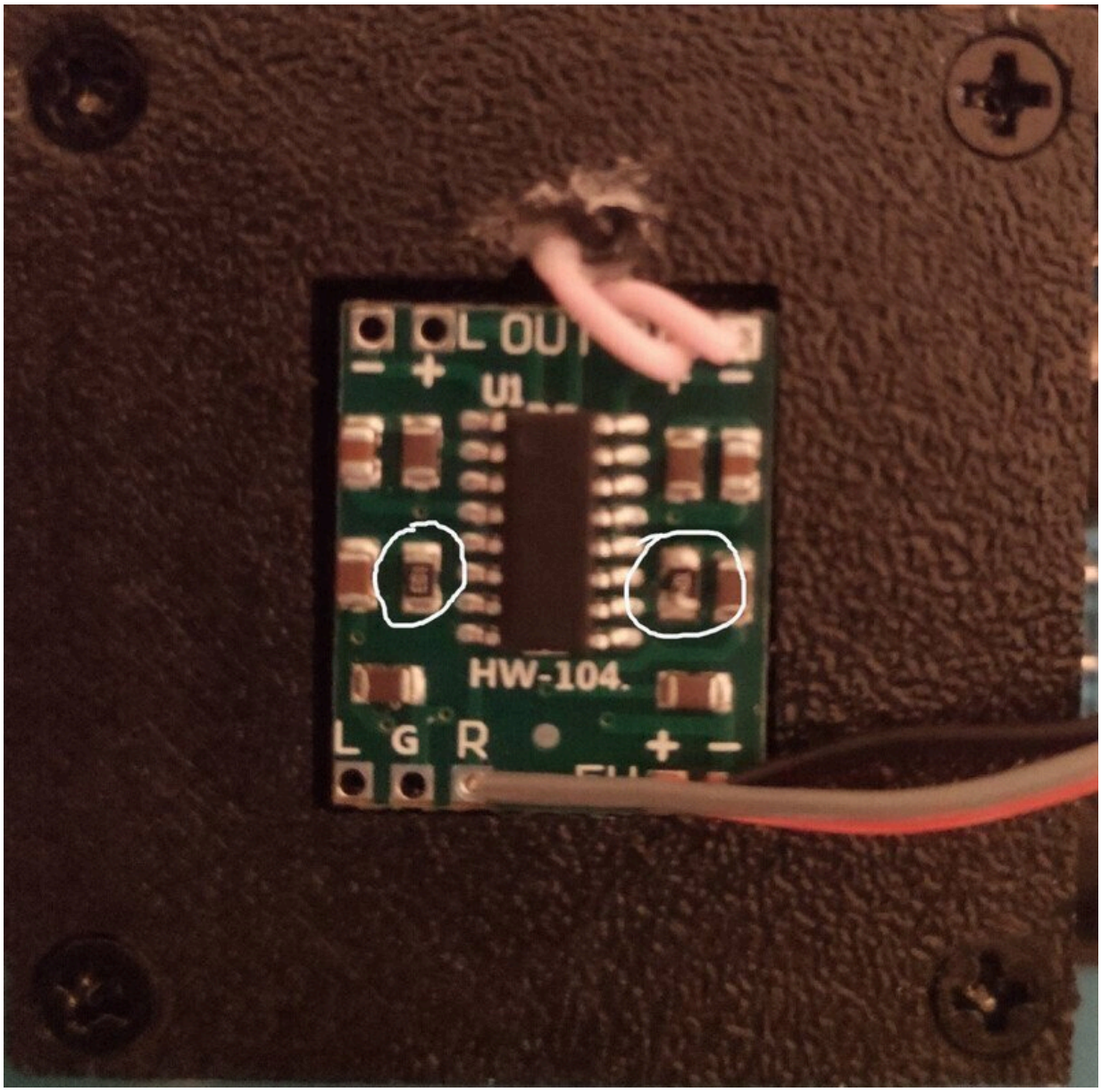

I have included a simple switch to isolate the amplifier when the speaker is not required, but this could also be achieved using a switch in the headphone jack.

There are numerous small speakers to choose from and the quality can be variable. In any case, a sealed enclosure can improve the quality of the sound. Speakers in this price range don't come with detailed specifications, usually just the impedance and power rating, so it isn't possible to design anything sophisticated. This page gives some simple rules of thumb.

I used some self-adhesive felt on the rear of the enclosure to help absorb echoes.

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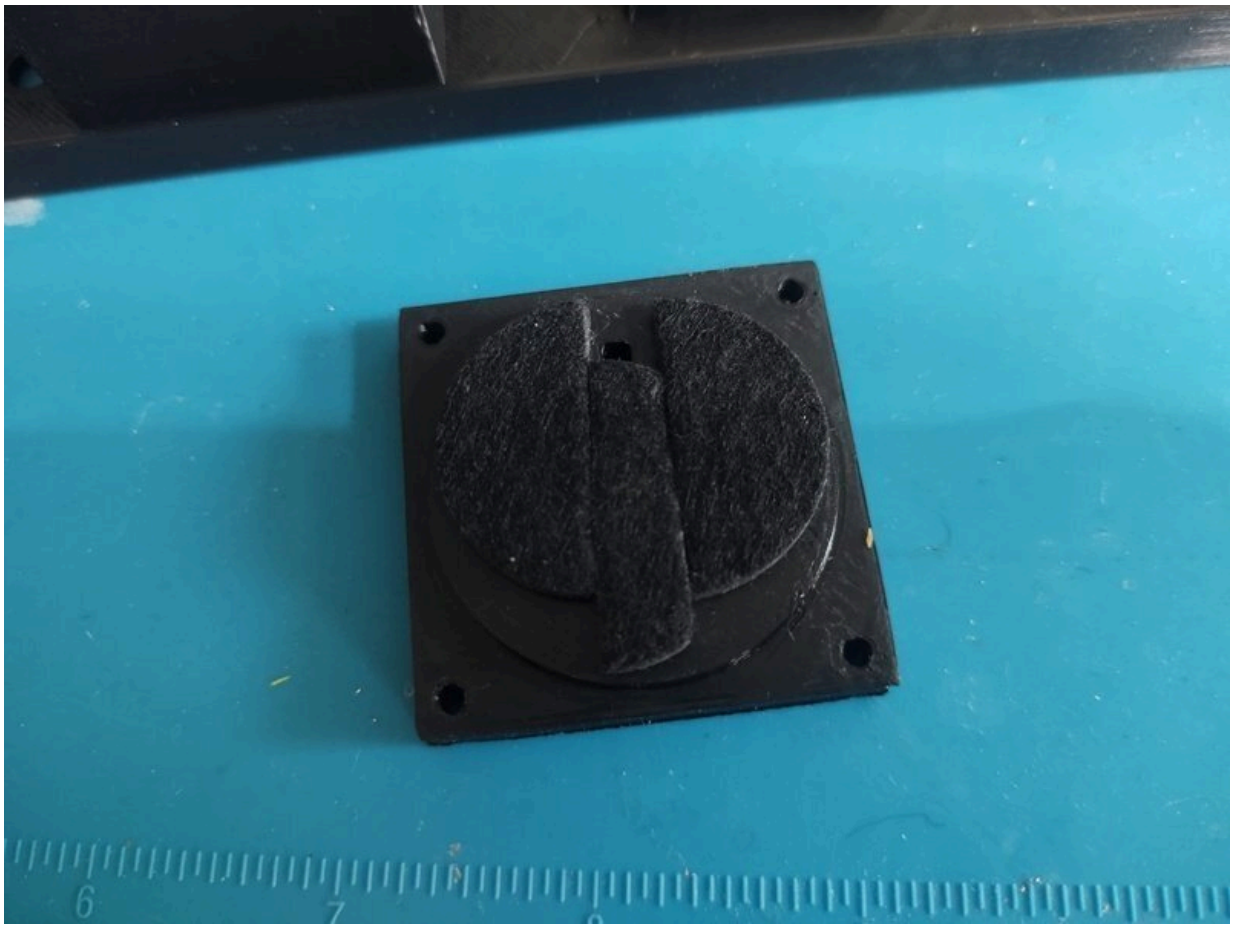

Conclusion

This SDR project continues to evolve, with numerous upgrades and improvements planned for the future.

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